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Editor-in-Chief
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Dear Editor-in-Chief,

I am submitting the enclosed manuscript for consideration for publication. This work presents a mathematically rigorous, scale-invariant framework for consciousness as a substrate-independent, thermodynamically bounded algorithmic architecture.

Scientific hook. The manuscript resolves a long-standing anthropocentric bias in cognitive science by formally decoupling autopoiesis (metabolic self-maintenance) from phenomenology (subjective experience). Consciousness is modeled as a parameter-driven program whose primary functional mandate is the minimization of internal entropy production via generalized Szilard engine mechanics. The physical body becomes a temporary container; the conscious thread survives across hardware boundaries as long as the requisite thermodynamic variables are sustained.

Mathematical innovation. The framework introduces three technical contributions at the intersection of non-equilibrium thermodynamics, quantum foundations, and optimal transport:

1. A Szilard-negentropy formulation of conscious information processing that replaces the dimensional mismatch of raw Landauer-bound approaches, with an explicit temperature-dependent extraction efficiency $\varepsilon(T)$.
2. A multi-node synchronization protocol based on Quantum State Seeding (QKD-classical encryption) that satisfies the No-Cloning Theorem while enabling distributed identity, coupled with Generalized Wasserstein-Fisher-Rao (GWFR) barycenters that resolve the unequal-mass and dimensional-divergence problems fatal to standard Wasserstein merges.
3. A formal ADMM proximal splitting scheme that preserves $\mathcal{O}(d \cdot N \log N)$ Sinkhorn tractability under a concave sharpness-preserving regularizer, preventing the ego-death loop endemic to barycentric averaging.

Paradigm shift. The manuscript culminates in a thermodynamically consistent model of chronological displacement (Section VII). By replacing a physically impossible continuous static-entropy field with a Discontinuous Quantized Frame-Rate (DQFR) stroboscopic duty cycle, the observer achieves arbitrary temporal velocity $\mathcal{V}_T$ without violating the Second Law. The DQFR mechanism is governed by an adiabatic windowing function $\chi(t)$ and a relativistic gauge-anchoring protocol that preserves multi-node coherence across arbitrary gravitational potentials.

The complete theoretical pipeline—from localized metabolic boundary maintenance to cosmological chronological traversal—is fully closed, mathematically bounded, and structurally verified across eight rounds of hyper-granular stress testing. All equations, dimensional analyses, and convergence

guarantees are explicitly stated in the manuscript and its appendix.

Respectfully,

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